

Ex1. $\log n! = n(\log n - 1)$

$$\log n! = \sum_{i=1}^n \log i \rightarrow \int_1^n \log x \, dx$$

$$x = e^y \quad dx = e^y dy \quad \begin{matrix} x=1 & x=n \\ y=0 & y=\log n \end{matrix}$$

$$\int_0^{\log n} y e^y dy = \left\{ y - \frac{dy}{e^y} \right\} \rightarrow e^y \left\{ \right.$$

$$= e^y \cdot y \Big|_0^{\log n} - \int_0^{\log n} e^y dy =$$

$$= n \log n - n + 1 \approx n(\log n - 1)$$

large n .

Ex2. $\langle n \rangle = \sum_n n \frac{(t)^n}{n!} e^{-t} = e^{-t} \frac{\partial}{\partial t} \sum_n \frac{(t)^n}{n!}$

derivation trick.

$$= e^{-t} \frac{\partial}{\partial t} [e^t] = e^{-t} \cdot e^t \cdot t = t$$

Ex3. $p(\vec{v}) d\vec{v} = A e^{-\beta \frac{mv^2}{2}} d\vec{v}$

3.1) $p(v)$? First we have to normalize

$$\int A e^{-\beta \frac{mv^2}{2}} dv_x dv_y dv_z$$

$$\int e^{-\frac{\beta m}{2}(v_x^2 + v_y^2 + v_z^2)} dv_x dv_y dv_z$$

$$= \pi \int_{\vec{v}(\vec{v})} dv_x e^{-\frac{\beta m}{2} v_x^2} A = \left(\sqrt{\frac{2\pi}{\beta m}} \right)^3 A = 1 \quad A = \left(\frac{2\pi}{\beta m} \right)^{3/2}$$

$$p(\vec{v}) = \left(\frac{2\pi}{\beta m} \right)^{-3/2} e^{-\frac{\beta m}{2} v^2}$$

$$p(v_x) = \int p(\vec{v}) dv_y dv_z = \sqrt{\frac{\beta m}{2\pi}} e^{-\frac{\beta}{2} v_x^2}$$

→ for each one $\sqrt{\frac{\beta}{2\pi}} \int_{-\infty}^{\infty} dv_y e^{-\frac{\beta}{2} v_y^2} = 1$

3.2) $\int_{-\infty}^{\infty} p(v_x) \cdot v_x dv_x = 0$ because it is symmetric around zero

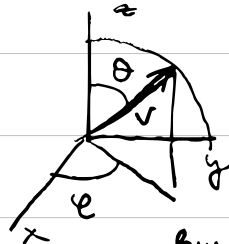
$\langle v_x \rangle = 0 = \text{mode}$

3) $p(v)$? we have to change to spherical coordinates (v, φ, θ)

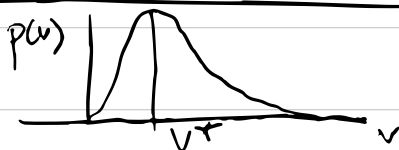
$$dv_x dv_y dv_z = v^2 dv d\varphi d\theta$$

$$\int_0^\pi \sin \varphi d\varphi \int_0^{2\pi} d\theta = 4\pi$$

2) 2π



$$p(v) = \int p(\vec{v}) v^2 d\theta \sin \varphi d\varphi = 4\pi v^2 e^{-\frac{\beta m}{2} v^2} \left(\frac{\beta m}{2\pi} \right)^{3/2}$$



34. Max value of v and $\langle v \rangle$

max value of v^2

$$f(v) \propto v^2 e^{-\frac{av^2}{2}} \quad \frac{d}{dv} f(v) = 2v e^{-\frac{av^2}{2}} - v^2 a v e^{-\frac{av^2}{2}} = 0$$

$$2v - av^3 = 0 \quad v=0$$

$$v = \sqrt{\frac{2}{a}}$$

$$v_{\max} = \sqrt{\frac{2}{\beta m}} = \sqrt{\frac{2k_B T}{m}}$$

$$\approx 1.414214 \frac{1}{\sqrt{\beta m}}$$

$$a = \beta m$$

$$\beta = \frac{1}{k_B T}$$

Only positive solution because $v \geq 0$.

$$\langle v \rangle = \frac{4\pi}{(2\pi)^{3/2}} (\beta m)^{3/2} \int_0^\infty dv v^3 e^{-\frac{\beta m}{2} v^2}$$

$$= \sqrt{\frac{2}{\pi}} (\beta m)^{3/2} \int_0^\infty dv v^3 e^{-\frac{\beta m}{2} v^2}$$

$$= \left\{ \begin{array}{l} v e^{-\frac{\beta m}{2} v^2} \rightarrow -\frac{1}{\beta m} e^{-\frac{\beta m}{2} v^2} \\ v^2 \rightarrow 2v dv \end{array} \right\}$$

$$= \sqrt{\frac{2}{\pi}} (\beta m)^{3/2} \left[-\frac{v^2}{\beta m} e^{-\frac{\beta m}{2} v^2} \right]_0^\infty + \frac{1}{\beta m} \int_0^\infty dv 2v e^{-\frac{\beta m}{2} v^2}$$

$$= \sqrt{\frac{2}{\pi}} (\beta m)^{3/2} \cdot \frac{2}{(\beta m)^2} = \sqrt{\frac{2}{\pi}} \frac{2}{\sqrt{\beta m}} \approx 1.596 \frac{1}{\sqrt{\beta m}}$$

For the modulus of the velocity

$\langle v \rangle \neq v_{\max}$ because the distribution is not symmetric around $v_{\max}$! (long tail to the right of $v_{\max}$)

3.5. Distrib for the linear momentum.

$$\vec{p} = m \vec{v} \quad dp_x = m dv_x \rightarrow dp_x dp_y dp_z = m^3 d\vec{v}$$

$$\boxed{p(\vec{p}) d\vec{p} = p(\vec{v}) d\vec{v}} \quad \begin{array}{l} \text{prob} \\ \text{density must be} \\ \text{conserved.} \end{array}$$

$$p(\vec{p}) m^3 d\vec{v} = p(\vec{v}) \quad p(\vec{p}) = \frac{1}{m^3} p(\vec{v})$$

$$\boxed{p(\vec{p}) = \sqrt{\frac{\beta}{2\pi m}} e^{-\frac{\beta}{2m} p^2}$$

3.6. $k = \frac{1}{2} m v^2 \quad dk = \frac{1}{2} m 2v dv$

$$p(k) dk = p(v) dv$$

$$p(k) m v dv = p(v) dv$$

$$p(k) = \frac{1}{m v} p(v) = \sqrt{\frac{2}{\pi}} \beta^{3/2} m^{1/2} \sqrt{\frac{k}{m}} e^{-\beta k}$$

$$v = \sqrt{\frac{2k}{m}} \quad \boxed{p(k) = \sqrt{\frac{2k}{\pi}} \beta^{3/2} e^{-\beta k}$$

4.

The marginal of a multi-variable distribution is the distribution we obtain for a random variable by integrating out over the other variables:

$$\text{Given } p(x, y), \quad p(x) = \int dy \, p(x, y)$$

↑
marginal

$$p(x, y) = Bx e^{-\frac{x^2}{2} - yx} \rightarrow \text{Normalizing constant}$$

$$p(x) = \int dy \, Bx e^{-\frac{x^2}{2} - yx}$$

$$\int dx \, dy \, Bx e^{-x^2/2} e^{-yx} = 1$$

$$\int_0^\infty dy \, e^{-yx} = \frac{1}{x} \quad ; \quad \int_0^\infty dx \, Bx \frac{1}{x} e^{-x^2/2} = \frac{B\sqrt{2\pi}}{2}$$

$$\frac{B\sqrt{2\pi}}{2} = 1 \quad B = \frac{2}{\sqrt{2\pi}}$$

$$p(x) = \frac{2}{\sqrt{2\pi}} \times e^{-\frac{x^2}{2}} \int_0^\infty dy \, e^{-yx} = \frac{2}{\sqrt{2\pi}} e^{-x^2/2}$$

$$\boxed{p(1) = e^{-1/2} \frac{2}{\sqrt{2\pi}}}$$

5. Convolution

Exponential distribution:

$$p(x) = \lambda e^{-\lambda x} \quad x \geq 0$$

$$p(x, y) = p(x) p(y) = \lambda_x \lambda_y e^{-\lambda_x x} e^{-\lambda_y y}$$

$$p(z) = \int dx dy \lambda_x \lambda_y e^{-\lambda_x x - \lambda_y y} \delta(z - (x+y))$$

$$= \int_0^z dx \lambda_x \lambda_y e^{-\lambda_x x - \lambda_y (z-x)}$$

$$= \lambda_x \lambda_y e^{-\lambda_y z} \int_0^z dx e^{-(\lambda_x - \lambda_y)x}$$

$$= \frac{\lambda_x \cdot \lambda_y}{\lambda_x - \lambda_y} e^{-\lambda_y z} [e^{(\lambda_x - \lambda_y)z} - 1]$$

$$\boxed{p(z) = \frac{\lambda_x \cdot \lambda_y}{\lambda_x - \lambda_y} [e^{-\lambda_x z} - e^{-\lambda_y z}]}$$

What happens if $\lambda_x = \lambda_y$?

$$\boxed{p(z) = \int_0^z e^{-\lambda z} \int_0^z dx = \underline{z \lambda^2 e^{-\lambda z}}}$$

6. Moments & Cumulants

$$p(x) = \frac{1}{\sqrt{2\pi}} e^{-\frac{1}{2}x^2} \quad p(y) = \frac{1}{\sqrt{2\pi}} e^{-\frac{1}{2}y^2}$$

$$p(x, y) = p(x) \cdot p(y)$$

$$p(z) = \int p(x) p(y) \delta(z - (x^2 + y^2)) dx dy$$

In spherical coordinates $dx dy = \rho d\rho d\phi$

$$\rho = \sqrt{x^2 + y^2}$$

$$p(z) = \int_0^{2\pi} d\phi \int_0^{\infty} d\rho \frac{e^{-\frac{\rho^2}{2}}}{2\pi} \rho \delta(z - \rho^2)$$

$$= \frac{2\pi}{2\pi} \int_0^{\infty} d\rho e^{-\frac{\rho^2}{2}} \rho \delta(z - \rho^2)$$

$$\delta(g(x)) = \frac{\delta(x - x_0)}{|g'(x_0)|} \quad \begin{matrix} \nearrow \text{roots of} \\ g(x) \\ \nwarrow \end{matrix}$$

$$\delta(z - \rho^2) = \frac{1}{2\sqrt{z}} \left[\delta(\rho - \sqrt{z}) + \delta(\rho + \sqrt{z}) \right] \quad \begin{matrix} \nearrow \text{It cannot be } \rho > 0 \\ \nwarrow \end{matrix}$$

$$p(z) = \int_0^{\infty} d\rho \frac{1}{2} \frac{\rho}{\sqrt{z}} e^{-\frac{\rho^2}{2}} [\delta(\rho - \sqrt{z})]$$

$$\underline{p(z)} = \frac{1}{2} \frac{\sqrt{z}}{\sqrt{z}} e^{-z/2} = \frac{1}{2} e^{-z/2}$$

$$p(z) = \frac{1}{2} e^{-z/2}$$

$$\langle z^n \rangle = \int dz \, z \frac{e^{-z/2}}{2}$$

Alternatively we can define the Moment Generating Function

$$M(a) = \langle e^{az} \rangle = \int_0^\infty dz \, \frac{e^{-z/2}}{2} e^{az} = \frac{1}{2(1/2 - a)}$$

$$\text{So that } \langle z^n \rangle = \left. \frac{d^n M(a)}{da^n} \right|_{a=0}$$

$$\langle z \rangle = \left. \frac{1}{2} \frac{1}{(1/2 - a)^2} \right|_{a=0} = 2$$

$$\begin{aligned} \langle z \rangle &= \int dz \, z \frac{e^{-z/2}}{2} = \int_0^\infty dz \, z e^{-z/2} = \left[-2 e^{-z/2} \right]_0^\infty = 2 \\ &= -2 e^{-z/2} \Big|_0^\infty + \int_0^\infty e^{-z/2} dz = -2 e^{-z/2} \Big|_0^\infty = 2 \end{aligned}$$

$$\langle z^2 \rangle = \left. \frac{1}{2} \frac{-2 \cdot -1}{(1/2 - a)^3} \right|_{a=0} = 8 \quad \langle z^3 \rangle = \left. \frac{3}{(1/2 - a)^4} \right|_{a=0} = 48$$

$$[C_1(z) = \langle z \rangle = 2] \quad [C_2 = \langle z^2 \rangle - \langle z \rangle^2 = 8 - 4 = 4]$$

$$\begin{aligned} [C_3(z) = \langle (z - \langle z \rangle)^3 \rangle] &= \langle z^3 - 3z^2 \langle z \rangle + 3 \langle z \rangle^2 z - \langle z \rangle^3 \rangle \\ &= \langle z^3 \rangle - 3 \langle z^2 \rangle \langle z \rangle + 3 \langle z \rangle^3 = 48 - 3 \cdot 8 \cdot 2 + 2 \cdot 2^3 = 24 \end{aligned}$$

Ex 7. MONOTONOUS $\leftarrow$ we remove all 0's except for one

$\rightarrow$ Num permutations $\frac{10!}{1! 2!}$ $\leftarrow$ repeated N's
 $\rightarrow$ # of permutations with all 0's together

It's like having a word with a single 0

MONTONUS 7 letters $\frac{7!}{2!} = 2520$
 repeated $\rightarrow$ 2! N's

30's + 10 separate.

MONTNUS
 $\uparrow \quad \uparrow \uparrow \uparrow \uparrow$

For each of the perms of MONTNUS,
 the extra 0 can go
 in any of the places signified
 by the arrow (6 total)

So the
 # of perms is $\frac{7!}{2!} \times 6 = 15120$

Ex 8. 700 misprints in 1400 pages.

Q.1 p one page contains no mistakes:

n mistakes 'm' in g pages 'p' $\rightarrow$ p's are boundaries between m's. Since there is always a last 'p', there are g+1 mobile 'p's

m m m | m m | m | | m m m 1
 p p p p p

permutations of $n+g+1$ elements $\frac{(n+g+1)!}{n! (g+1)!} \rightarrow$ identical m's $\rightarrow$ identical pages

→ # configs with a page mistake free
 n mistakes distributed in $g-1$ pages

$$\frac{(n+g-2)!}{n! (g-2)!}$$

$$p(\text{page no m's}) = \frac{(n+g-2)!}{n! (g-2)!} \frac{n! (g-1)!}{(n+g-1)!}$$

$$= \frac{g-1}{n+g-1} = \frac{1400-1}{2100-1} \approx 0.67$$

8.2 → p. One page contains 2 mistakes.

I have $n-2$ mistakes in $g-1$ pages

$$p(1 \text{ page } 2 \text{ m's}) = \frac{(n+g-3)!}{(n-2)! (g-2)!} \frac{n! (g-1)!}{(n+g-1)!}$$

$$= \frac{n(n-1)(g-1)}{(n+g-1)(n+g-2)} = 0.07 (< 1-p(0))$$

EX 9. Loaded die: $246 \rightarrow 3 \times 135$

$$\text{so } p(2) = p(4) = p(6) = 3 p(1)$$

$$\sum p_i = 1 = 12 p(1) \quad p(1) = \frac{1}{12} \quad P_{\text{odd}} = 3 p(1) = \frac{1}{4}$$

9.1 Binomial distribution $\binom{N}{n} p^n (1-p)^{N-n}$

$$p(3 \text{ odd out of } 12) = p_{\text{odd}}^3 (1-p_{\text{odd}})^{12-3} \binom{12}{3} = 0.2881$$

$$p(30 \text{ odd out of } 120) = \binom{120}{30} p_{\text{odd}}^{30} (1-p_{\text{odd}})^{90} = 0.08325$$

9.2 Gaussian distribution

$$p(x) = \frac{1}{\sqrt{2\pi\sigma^2}} e^{-\frac{(x-\bar{x})^2}{2\sigma^2}}$$

$\bar{x} = \# \text{ of expected odd}$

$$= p_{\text{odd}} \cdot N \text{ rolls}$$

$$\sigma^2 = p_{\text{odd}}(1-p_{\text{odd}})N$$

$$\begin{aligned} \sum_n \binom{N}{n} p^n (1-p)^{N-n} e^{na} &= \sum_n (pe^a)^n q^{N-n} \binom{N}{n} \\ &= \sum_n (pe^a)^n q^{N-n} \binom{N}{n} = (q + pe^a)^N \end{aligned}$$

$$\begin{aligned} \langle n \rangle &= \left. \frac{d}{da} (q + pe^a)^N \right|_{a=0} = N (q + pe^a)^{N-1} pe^a \Big|_{a=0} \\ &= N \cdot p \end{aligned}$$

$$\begin{aligned} \langle n^2 \rangle &= \left. \frac{d^2}{da^2} (q + pe^a)^N \right|_{a=0} = N pe^a (q + pe^a)^{N-1} \\ &\quad + N(N-1) p^2 e^{2a} (q + pe^a)^{N-2} \Big|_{a=0} \\ &= Np + N(N-1)p^2 \end{aligned}$$

$$\begin{aligned} \sigma^2 &= \langle n^2 \rangle - \langle n \rangle^2 = Np + N(N-1)p^2 - N^2 p^2 \\ &= Np + N^2 p^2 - Np^2 - N^2 p^2 \\ &= Np(1-p) = Npq. \end{aligned}$$

$$\bar{x}_{\text{odd}} = 12 \cdot \frac{1}{4} = 3$$

$$N=12 \quad \sigma_{\text{odd}}^2 = 12 \cdot \frac{1}{4} \cdot \frac{3}{4} = 9/4 = 2.25 \quad p(3)_{\text{Gauss}} = 0.266 \quad (\text{vs } 0.2881)$$

$$N=120 \quad \bar{x}_{\text{odd}} = 120 \cdot \frac{1}{4} = 30$$

$$\sigma_{\text{odd}}^2 = 120 \cdot \frac{1}{4} \cdot \frac{3}{4} = \frac{90}{4} = 22.5 \quad p(30)_{\text{Gauss}} = 0.0841 \quad (\text{vs } 0.08385)$$

EX10. 3-faced dice d_1 & d_2

Distinguishable $\rightarrow$ All legal

Bosons $\rightarrow$ legal if $d_2 \geq d_1$

Fermions $\rightarrow$ legal if $d_2 > d_1$. sum 5.

10.1 $p(4)$ — (D) Distinguishable — $\frac{3}{9} = \frac{1}{3}$ possible ^{— 133122}

$d_1 \backslash d_2$	1	2	3
1	DB	DBF	DBF
2	DB	DB	DBF
3	DB	DB	DB

— (B) Bosons = $\frac{2}{6} = \frac{1}{3}$ possible ^{1122 132333}

— (F) Fermions = $\frac{1}{3}$ possible ^{12 1323}

10.2 $p(d_1 = d_2)$ — Distinguishable $\frac{3}{9} = \frac{1}{3}$?

— Bosons $\frac{3}{6} = \frac{1}{2}$

— Fermions 0

10.3 triple (three dice equal)

Distinguishable : $\frac{3}{3 \cdot 3 \cdot 3} = \frac{1}{9}$

Bosons = $\frac{3}{10}$

$d_1 \boxed{1} \rightarrow 6$ $d_3 > d_2 > d_1 \rightarrow 10$

V of a hypercube $L=3$

$d_2 2 \rightarrow 3$ $\begin{smallmatrix} 2223 \\ 33 \end{smallmatrix}$

$d_3 3 \rightarrow 1 (333)$

enhancement = $\frac{3/10}{1/9} = \frac{27}{10} = 2.7$

In general for M rolls $\rightarrow p(M\text{-tuple})$

$p(\text{Bosons } M\text{-tuple}) = \frac{3}{(M+1)(M/2+1)}$ $\begin{smallmatrix} n & m & p \\ 1 & 2 & 3 \end{smallmatrix} \rightarrow \underline{n+m+p=M}$

$p(\text{dist } M\text{-tuple}) = \frac{3}{3^M}$

All legal rolls have $\begin{smallmatrix} n_1 & n_2 & M-(n_1+n_2) \\ 111 & 222 & 333 \end{smallmatrix}$

$$\begin{aligned} \sum_{n_1=0}^M \sum_{n_2=0}^{M-n_1} 1 &= \sum_{n_1=0}^M (M-n_1+1) \\ &= (M+1)(M+1) - \sum_{n_1=0}^M n_1 \\ &= (M+1)^2 - \frac{M+1}{2} \cdot M \\ &= (M+1)\left(\frac{M}{2}+1\right) \end{aligned}$$

So that the enhancement for M -tuples is

$$\frac{p(B, \text{tuple})}{p(D, \text{tuple})} = \frac{3^{1/(M+1) \times (\frac{M}{2} + 1)}}{3/3^M} = \frac{3^M}{(M+1)(\frac{M}{2} + 1)}$$

Ex. 11.

11.1 Distinguishable: In this case each part has $p = \frac{1}{N}$ of being in one box so each config of balls in boxes has

$p(\text{config}) = \frac{1}{N^n}$. The number of favorable configurations are those in which I put the first ball in one of the N boxes, the second in one of the $N-1$ remaining boxes and so on $N(N-1) \dots (N-n+1)$ (Note $(N-n+1)$ for the last ball)

$$\text{So } p(\text{at most 1 ball per box}) = \frac{\frac{N!}{(N-n)!}}{N^n}$$

11.2 Particles are identical and have no restrictions:

The # of possible configurations is smaller than for the previous case.

- A way to count this situation is to visualize boxes as boundaries between balls

$\underbrace{000}_1 \mid 000 \underbrace{00}_{N-1} \mid 0 \mid 1 \mid 000 \mid 111 \mid 00 \mid 0 \mid$
 n balls N-1 boundaries the one at the end is fixed

$$\# \text{ possible configs } \frac{(n+N-1)!}{\underbrace{n!(N-1)!}_{\text{balls and boxes identical}}}$$

$$\# \text{ favorable configs } C\left(\begin{matrix} N \\ n \end{matrix}\right) = \frac{N!}{n!(N-n)!}$$

$$P_{BE} = \frac{N!}{n!(N-n)!} \cdot \frac{n!(N-1)!}{(n+N-1)!} = \frac{N!(N-1)!}{(n-n)!(n+N-1)!}$$

11.3 Fermions: 2 balls cannot be in the same box

$\boxed{P_F = 1}$ All possible configurations are favorable configurations

Note that:

$$P_F > P_{\text{classical}} > P_{BE}$$

$\rightarrow$ This type of particles tend to group more than classical ones!